Quiz 3: Monday, regular quiz

Quiz 4: Thursday, partner activity, no laptops, but open notes

Agenda

Cubic Splines

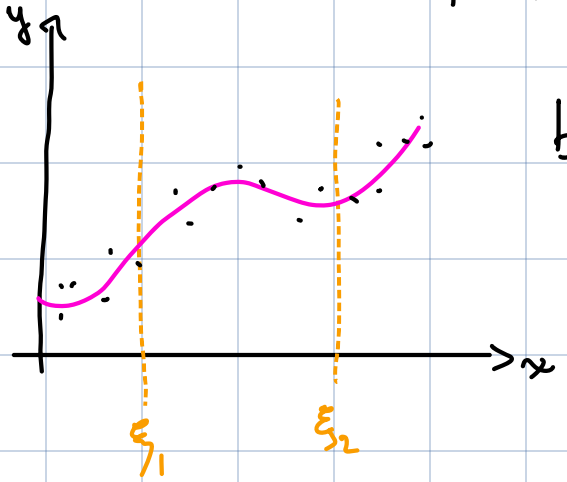
Smoothing Splines

1. Cubic Splines

1.1 Review

$$y_i = \beta_0 + \sum_{j=1}^{K+3} \beta_j b_j(x_i) + \varepsilon_i$$

If $K = 2$, we have $\beta_0, \beta_1, \ldots, \beta_5$



$b_1 = x_i$, $b_2 = x_i^2$ for example

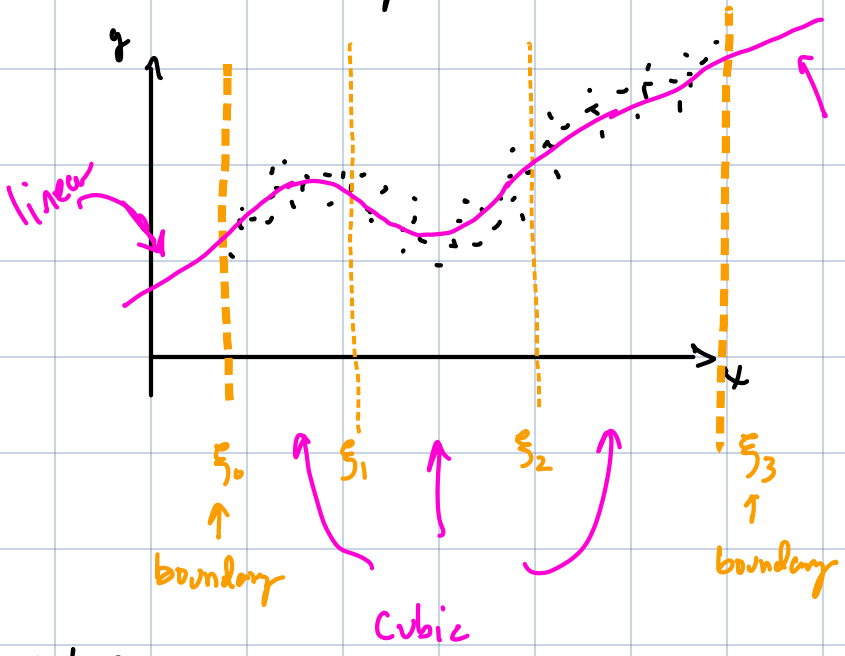
$b_3 = x_i^3$

b_4, b_5 can be truncated power basis:

$$h(x,\xi) = (x-\xi)_+^3 = \begin{cases} (x-\xi)^3 & x > \xi \\ 0 & \text{else} \end{cases}$$

1.3 Natural Cubic Spline

One more constraint: curve is linear beyond boundary knots ξ_0 and ξ_{k+1}



Why? Less extreme tails

unconstrained d.f. $2 + 4 + 4 + 4 + 2 = 16$

Knot Constraints $3 \quad 3 \quad 3 \quad 3 = 12$

c.d.f. $= 16 - 12 = 4$

or d.f. $= (K+2) + 4 - (2+2)$

- $(K+2)$: inner knots + boundary knots
- 4: boundary areas
- $(2+2)$: use 1 and x, not x^2 and x^3

or d.f. $= K+2$ ← interior knots only

$$N3 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & x & b_2 & b_3 \\ 1 & 1 & 1 & 1 \end{pmatrix}_{n \times 4}$$

complex

1.4 Quick Notes

- $x, x^2, x^3, \ldots, (x-\xi_1)^3, \ldots$ correlated, numerically unstable

  solutions:
  - + standardize x
  - + use orthonormal columns. (QR decomp)
  - + B-spline basis

- Choosing K - use C.V.
- Knot placement - by default, use quantiles

Round 1

12
6
4

Round 2

Y
N
N

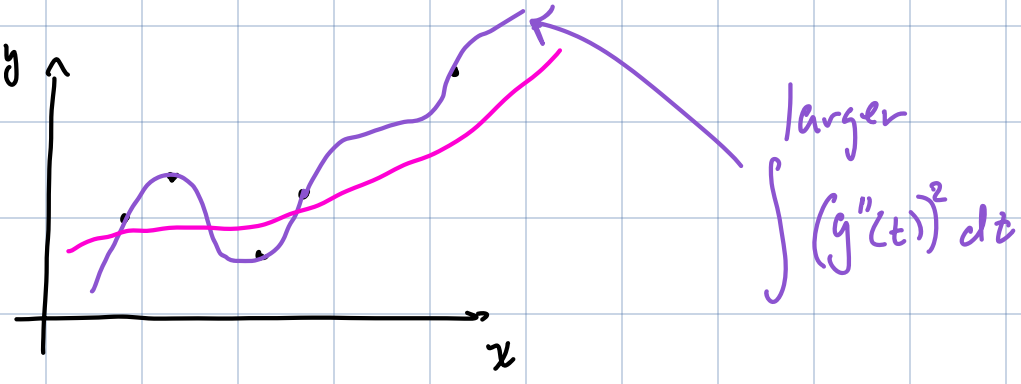
Round 3

C
A
B

2. Smoothing Splines

2.1 The criterion

$$\hat{g}_\lambda = \underset{g}{\operatorname{argmin}} \sum_{i=1}^{n} (y_i - g(x_i))^2 + \lambda \int \left(g''(t)\right)^2 dt$$



$\lambda = 0 \Rightarrow$ as wiggly as we want

$\lambda \to \infty \Rightarrow$ OLS line

Bias? Variance?

2.2 Solution

Given - ESL Ex 5.7: For $\lambda > 0$ and $n \geq 2$ distinct x_i, the g that minimizes the criterion is a natural cubic spline with knots at all $x_1, \ldots, x_n$.

Recall: For $x_1 < x_2 < \ldots x_n$, x_1 and x_n are boundary knots

3.1 Fitting

$$g(x) = \beta_0 + \sum_{j=1}^{n-1} \beta_j b_j(x)$$

$$\mathbb{B} = \begin{pmatrix} | & | & & | \\ 1 & x & \cdots & b_{n-1} \\ | & | & & | \end{pmatrix}_{n \times n}$$

Ex:

$$\mathbb{B} = \begin{pmatrix} x & b_2 & b_3 \\ 1 & 0 & 0 & 0 \\ 1 & 1 & 1/3 & 0 \\ 1 & 2 & 8/3 & 1/2 \\ 1 & 3 & 8 & 3 \end{pmatrix}_{4 \times 4}$$

The loss:

$$\sum_{i=1}^{n} (y_i - g(x_i))^2 = \| y - \mathbb{B}\beta \|_2^2 = (y - \mathbb{B}\beta)^T (y - \mathbb{B}\beta)$$

Penalty: $\int (g''(t))^2 dt = \int \left(\sum_{j=1}^{n-1} \beta_j b_j''(t) \right)^2 dt$

$$(\beta_1 b_1 + \beta_2 b_2)^2$$

$$(\beta_1 b_1)^2 + 2\beta_1 b_1 \beta_2 b_2 + (\beta_2 b_2)^2$$

$$= \int \sum_j \sum_k \beta_j \beta_k b_j''(t) b_k''(t) \, dt$$

$$= \sum_j \sum_k \beta_j \beta_k \int b_j''(t) b_k''(t) \, dt$$

$$= \beta^T \Omega \beta$$

$\Omega_{n \times n}$: symmetric, positive semi-definite, eig ≥ 0

$$(y - B\beta)^T (y - B\beta) + \lambda \beta^T \Omega \beta$$

$$RSS_\lambda(\beta) = \| y - B\beta \|_2^2 + \lambda \beta^T \Omega \beta$$

Note: $B^T B$ pos. def. eig > 0

$$\Rightarrow \hat{\beta} = \left(B^T B + \lambda \Omega\right)^{-1} B^T y$$

Looks like ridge regression

$$\hat{g}_\lambda = B\hat{\beta} = \underbrace{B\left(B^T B + \lambda \Omega\right)^{-1} B^T}_{S_\lambda} y$$

S_λ linear smoother matrix

$$= S_\lambda y \quad \text{linear function of } y$$

Check it:

If $\lambda = 0$ $\quad S_0 = B(B^T B)^{-1} B^T = B B^{-1} (B^T)^{-1} B^T = I$

B is square, invertible